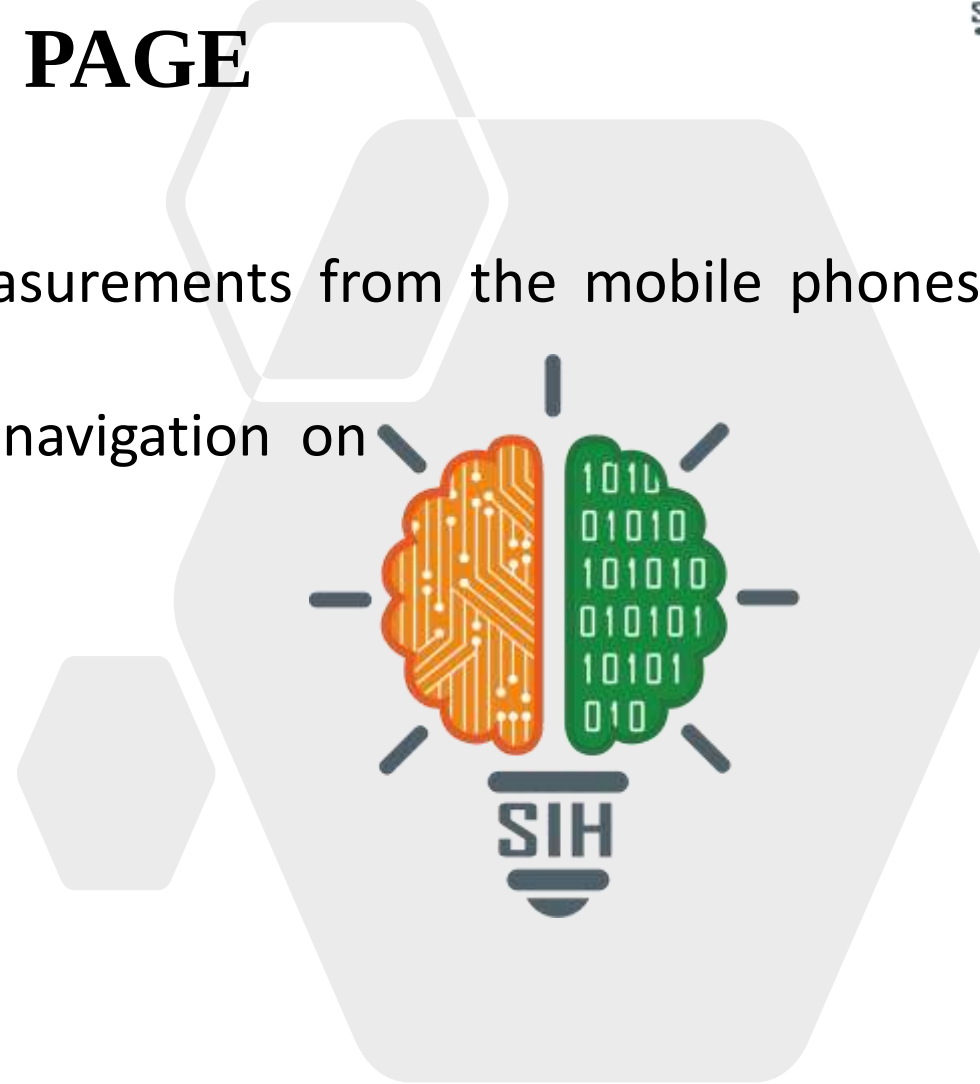


SMART INDIA HACKATHON 2025



TITLE PAGE

- **Problem Statement ID** –SIH25177
- **Problem Statement Title**-Use of measurements from the mobile phones (low cost preferred) to provide a safe autonomous navigation on the roads.
- **Theme**- Smart Vehicles
- **PS Category**- Software
- **Team ID**-112
- **Team Name**:-Team No Semicolon



IDEA TITLE

IDEA / SOLUTION :

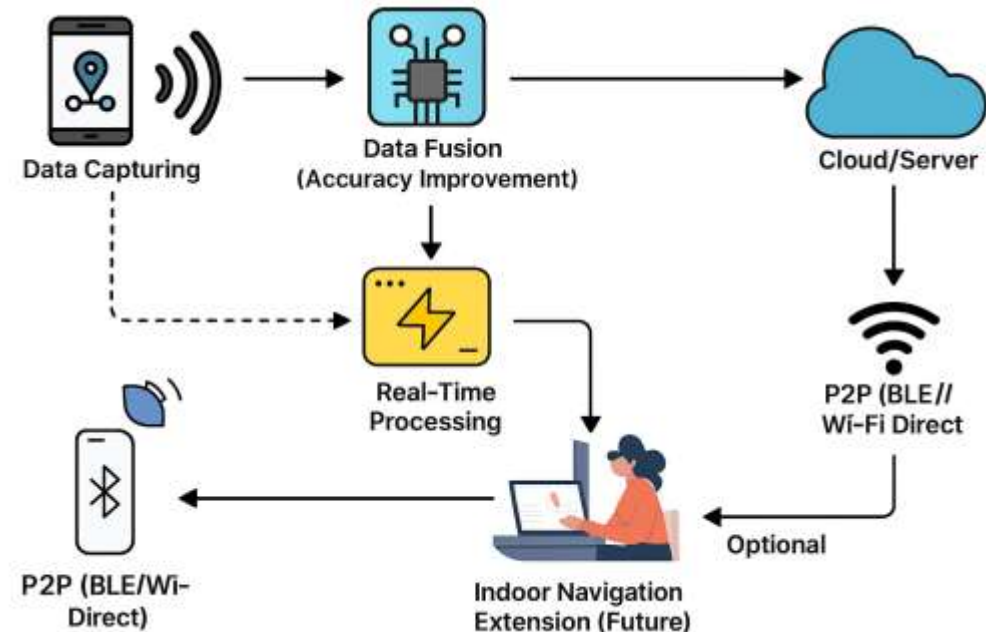
- Low-cost navigation using **phone sensors** instead of expensive hardware like LIDAR.
- **Collision detection** by data sharing among mobiles for better positioning .
- **Real-time alerts** for safe speed and distance control with **Indoor support** with static phones as Wi-Fi/BLE base stations.

PROBLEM RESOLUTION

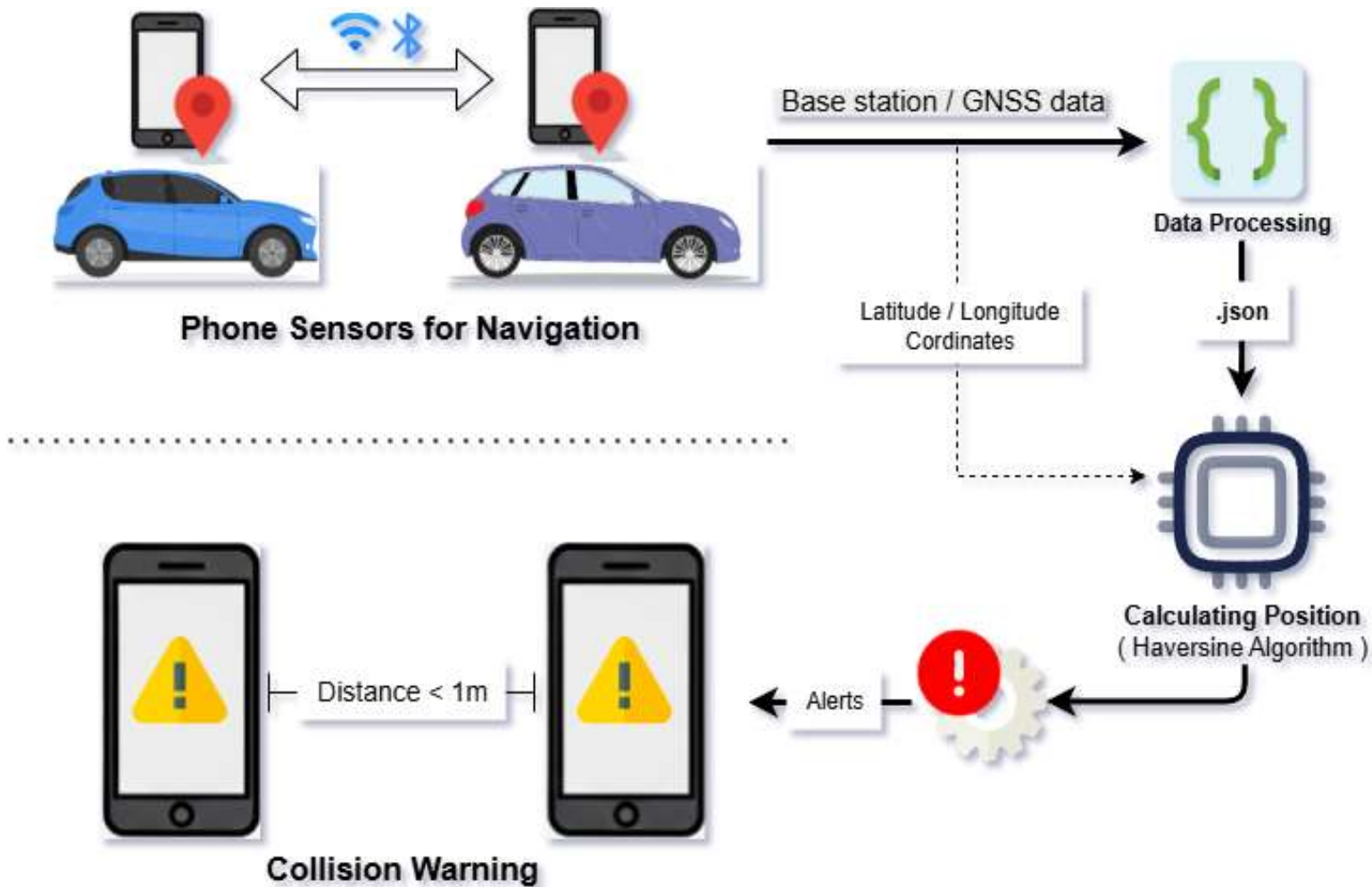
- **Improves accuracy** by combining GNSS & sensor data.
- WebSocket connection **Minimizes latency**.
- Avoids centralized failure by enabling distributed communication.

INNOVATION AND UNIQUENESS:

- **Peer-to-peer communication** reduces latency & server dependency.
- (GNSS + IMU) improve accuracy.
- **Scalable design** for both outdoor and indoor navigation.



TECHNICAL APPROACH



Technology Stack:

➤ Frontend:-



➤ Backend:-



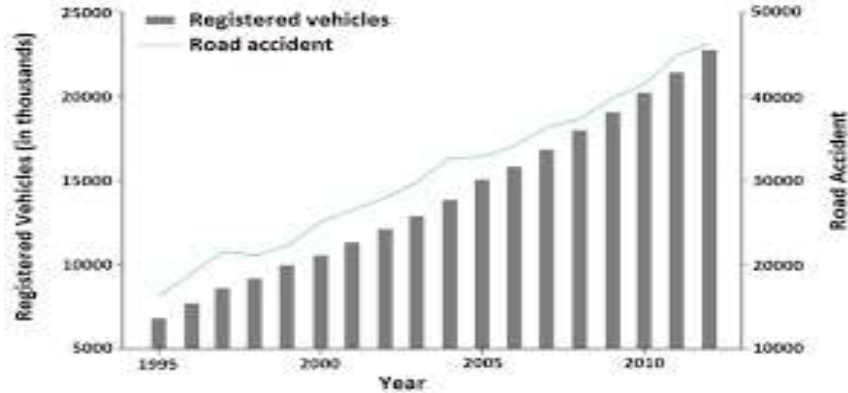
➤ Database:-



➤ Development Tools:-



FEASIBILITY AND VIABILITY



FEASIBILITY ANALYSIS:-

- **Technical** – Proven tech stack with GNSS and WebSocket-ready backend.
- **Financial** – Low-cost setup using open-source tools and cloud scaling.
- **Market Fit** – High demand for affordable collision-prevention solutions.
- **Operational** – Easy deployment with mobile devices and web access.



VIABILITY ANALYSIS:-

- **Technical** – The system allows future enhancements like AI-based data fusion and peer-to-peer communication.
- **Financial** – Low development and infrastructure costs make the project sustainable for long-term deployment across multiple fleets and cities.
- **Market Fit** – Strong adoption potential exists in fleet management, emergency services, and smart city traffic systems due to increasing safety demands.
- **Operational** – The platform can seamlessly integrate into existing transport operations while supporting multi-device tracking and real-time monitoring.

CATEGORY	POTENTIAL IMPACT	BENEFITS OF THE SOLUTION
High accuracy positioning	Data Fusion: combine low cost raw GNSS data(GPS) from the mobile phone with IMU sensors	Makes autonomous navigation more affordable & accessible
Base Station (mobile phones)	Used as local anchors in areas where satellite signals fail (like indoors) peer to peer communication is used.	Expanded coverage: ensures the navigation and warning system works everywhere.
Data sharing (real time)	Mobile phones constantly exchange their position/speed data with each other.	Enables immediate detection of proximity/collision, ensuring timely warning and faster reaction time
Driving control	Algorithm governs speed & safe distance between vehicles	Ensures smoother traffic flow and reduces chances of collision.

RESEARCH AND REFERENCES



Github Repo:- <https://github.com/GurukantPatil01/gnss-phone-collision-warning.git>

Deployed Link :- gnss-phone-collision-warning.vercel.app

[1] *Android Raw GNSS Measurement Datasets for Precise Positioning* — Frank van Diggelen, Mohammed Khider : <https://research.google/pubs/android-raw-gnss-measurement-datasets-for-precise-positioning/>

[2a] Haversine Algorithm Design using the Google Maps API Method for Android-based Public Security Applications — <https://mail.ijcttjournal.org/archives/ijctt-v69i2p108>

[3] Survey of smartphone-based sensing in vehicles for intelligent transportation system applications :<https://ietresearch.onlinelibrary.wiley.com/doi/full/10.1049/iet-its.2014.0248?>